

Sanhitha Varma Konduru

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Professional Summary

Results-driven Data Engineer with 3 years of experience in designing, building, and optimizing scalable data pipelines and analytics solutions across industries like finance. Proficient in leveraging big data technologies. Skilled in developing and managing ETL processes, real-time data streaming, data warehousing, and machine learning integration using tools such as Apache Airflow, Databricks, Snowflake, and Azure Synapse Analytics. Adept at writing optimized SQL queries and managing relational databases like MySQL and PostgreSQL for efficient data storage and retrieval.

Education

University of North Texas

Master in Computer Science

Denton, TX

Aug 2022 – May 2024

Andhra University

Bachelor in Computer Science

Visakhapatnam, India

July 2017 – May 2021

Work Experience

Data Engineer, S&P Global

Aug 2023 – Present | Remote, USA

- Designed and managed scalable data pipelines for the ingestion, transformation, and storage of large datasets, improving data accessibility for analytics and reporting purposes.
- Collaborated with cross-functional teams to align data solutions with S&P Global's objectives, improving data-driven decision-making.
- Designed and optimized ETL processes using Apache Airflow and Talend, achieving a 30% reduction in processing time.
- Utilized Azure to build and deploy a data lake, enhancing data storage and retrieval capabilities for diverse sources.
- Engaged in data governance initiatives to ensure compliance with data privacy regulations, contributing to data security and integrity.
- Automated data ingestion and transformation with Azure Data Factory and Apache Airflow, ensuring data consistency and reducing manual effort.
- Performed data transformations using Databricks, leveraging Apache Spark to efficiently handle large financial datasets for analytics.
- Developed predictive models using machine learning algorithms to forecast market trends, supporting investment decision-making.
- Created technical documentation in Jira for data pipelines and ETL processes, facilitating team collaboration and onboarding.
- Implemented data flows with Apache NiFi to automate financial data processing, ensuring real-time availability and operational efficiency.
- Managed integrated data solutions using Azure Synapse Analytics, enhancing reporting capabilities through seamless data ingestion and transformation.
- Designed and implemented Power BI reports and dashboards, transforming complex data sets into actionable insights, which enhanced visibility into key performance metrics and supported strategic decision-making.
- Designed and implemented Star Schema data models to optimize query performance and facilitate efficient reporting, enhancing data retrieval speed for analytical processes.

Data Engineer, Accenture

Feb 2020 – June 2022 | Visakhapatnam, India

- Developed and maintained SQL scripts for efficient ETL processes, optimizing performance and improving data retrieval times.
- Automated data processing tasks using Python and Pandas, handling large datasets efficiently and reducing manual workload by 20%.
- Assisted in building and deploying data pipelines with Apache Airflow, ensuring reliable and timely data flow across systems.
- Engaged in data modeling with Lucidchart, designing scalable data structures to support business analytics needs.
- Performed data integration using RESTful APIs, facilitating seamless data collection from external sources.
- Leveraged AWS S3 and AWS Glue for data storage and transformation, enhancing data accessibility and scalability.
- Participated in performance tuning of data pipelines, identifying bottlenecks and implementing optimizations to improve processing efficiency.
- Utilized Snowflake architecture to create and manage data warehouses, enabling scalable storage solutions and improving data querying capabilities across multiple business units.
- Utilized Git for version control, effectively managing code changes and collaborating with team members.
- Engaged in data validation and quality assurance, using profiling techniques to identify and rectify inconsistencies in datasets.
- Designed CI/CD pipelines for data workflows with Jenkins and Azure DevOps, automating deployments and reducing deployment times by 30%.
- Developed and optimized data processing workflows in Azure Databricks, leveraging Apache Spark to efficiently handle large-scale data transformations and enhance overall analytics performance.

- Developed and maintained interactive dashboards in Tableau, enabling stakeholders to visualize key metrics and insights from large datasets, which facilitated data-driven decision-making across the organization.

Technical Skills

Programming Languages: Python, Java, Scala, SQL

Data Warehousing: Redshift, Big Query, Snowflake, Teradata

Database Management: MySQL, PostgreSQL, MongoDB, Cassandra

ETL Tools: Apache NiFi, Talend, Informatica, Airflow

Cloud Platforms: AWS, Azure

Containerization: Docker, Kubernetes

Data Visualization: Tableau, Power BI

Big Data Technologies: Hadoop, Apache Spark, Apache Hive, Apache HBase, Apache Kafka, Apache Flink, Apache Pig, MapReduce, Apache Storm

Projects:

Data processing and information retrieval of atmospheric measurements using fuzzy's rule

- Implemented fuzzy logic rules for data cleaning and handling large datasets, removing NULL values.
- Utilized machine learning algorithms to forecast relationships and linearity among variables.
- Applied predictive modeling techniques for data-driven insights and pattern recognition.

Driver drowsiness detection using haarcascade classifiers

- Developed a real-time driver fatigue monitoring system using OpenCV and Python, enabling early detection of drowsiness to prevent over 100+ potential accidents annually.
- Implemented advanced image processing techniques, including face and eye detection using Haar Cascade classifiers, accurately identifying driver drowsiness with 95% precision.
- Integrated a non-intrusive computer vision solution, leveraging webcam input to monitor drivers' eye movements without disrupting their focus, enhancing safety of commuters.
- Optimized the system's performance by utilizing OpenCV's computational efficiency, achieving real-time processing of 30+ frames per second on a standard laptop configuration.

Social Media Sentiment Analysis

- Created a sentiment analysis pipeline for social media data, utilizing Apache Spark and Databricks to process and analyze large volumes of tweets. Developed machine learning models in Python to classify sentiment, and visualized results in Power BI, providing insights into public perception and engagement for marketing strategies.

Certificates

Microsoft Certified: Azure AI Fundamentals

Microsoft Certified: Azure Data Fundamentals

Technical Support Fundamentals by Google